

EXAM 7 – SPRING 2013

20. (5 points)

Given the following 2013 information for an insurance company:

Gross premiums	\$250 million
Gross loss ratio	60%
Gross expense ratio	32%
Risk-free interest rate	1.5%
Investment yield	3.5%
Beginning surplus	\$1 billion
Probability of distress without reinsurance	2%

The insurance company is considering a reinsurance treaty with the following characteristics:

Proposed reinsurance	20% quota share
Ceding commission	10% of reinsurance premiums
Probability of distress with reinsurance	0.4%

- All premiums, expenses and commissions are collected or paid at the beginning of the year.
- All claims are paid at the end of the year.
- The insurance portfolio composition is not expected to grow or change in future years.

a. (4 points)

Using the risk-adjusted present value of future earnings, quantify the effect on the firm's value of purchasing the quota share reinsurance.

b. (1 point)

Identify and briefly describe two alternative approaches to quantifying the value of risk transfer to the insurance company.

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Question 20 Sample Answer

Solution 1

a) Without Reinsurance:

$$D = \frac{1-d}{1+r} = \frac{1-0.02}{1+0.015} = 0.9655$$

$$Value = Earnings \times \frac{D}{1-D}$$

$$UW \text{ Earnings} = 250M - (0.6+0.32) \times 250M = 20M$$

$$Investment = (1B + 250M - 0.32 \times 250M) \times 0.035 = 40.95M$$

$$Value = (20M + 40.95M) \times \frac{0.966}{1-0.966} = 1705.7M$$

With Reinsurance:

$$D = \frac{1-d}{1+r} = \frac{1-0.004}{1+0.015} = 0.9813$$

$$UW \text{ Earnings} = 250M \times 0.8 - 250M \times 0.8 \times 0.6 - 250M \times 0.32 + 250M \times 0.1 \times 0.2 = 5M$$

$$Investment = (1B + 250M \times 0.8 - 0.32 \times 250M + 250 \times 0.1 \times 0.2) \times 0.035 = 39.375M$$

$$Value = (5M + 39.375M) \times \frac{0.9813}{1-0.9813} = 2328.6M$$

The value increases by $2328.6M - 1705.7M = 622.9M$

b)

(1) Efficient Frontier Graph – which plot U/W profit against different risk measure such as, VaR, or TvaR. The reinsurance option in the upper left corner is preferred since it has higher U/W profit with low risk

(2) Cost Allocation Method – holding capital is not free, which required a return of capital. We compare the total cost of holding capital and reinsurance net cost, to see which option has lower cost

Solution 2

a) $E \times M = E \times \frac{D}{1-D}$ where $D = \frac{1-d}{1+r}$

Without Reinsurance:

$$E = UW \text{ income} + Investment \text{ Income}$$

$$E = 250 - 0.92 \times 250 + (1000 + 250 \times (1 - 0.32)) \times 0.035 = 20 + 1170 \times 0.035 = 60.95$$

$$D = \frac{1-2\%}{1+1.5\%} = 0.965517$$

$$\text{Risk adjusted PV (Earnings)} = 60.95 \times \frac{0.965517}{1-0.965517} = 1706.6$$

With Reinsurance:

$$\text{UW income} = 0.8 \times 250 - 0.8 \times 0.6 \times 250 - 0.32 \times 250 + 0.1 \times 0.2 \times 250 = 5$$

$$\text{Investment Income} = (1000 + 0.8 \times 250 - 0.32 \times 250 + 0.1 \times 0.2 \times 250) \times 3.5\% = 1125 \times 0.035 = 39.375$$

$$D = \frac{1-0.4\%}{1+1.5\%} = 0.98128$$

$$\text{Risk adjusted PV (Earnings)} = (5 + 39.375) \times \frac{0.98128}{1-0.98128} = 2326.18421$$

$$\text{Difference: Impact of RI} = 2326.18421 - 1706.6 = 619.58421 \text{ M}$$

b)

- (1) Simple Factor: Value = prob of distress $\times$ Equity
Difference with or without reinsurance quantifies the risk transfer
- (2) Efficient Frontier Graph: the expected U/W profit against a risk measure. More U/W profit expected should be associated with a higher risk measures. The closest to the efficient frontier of the program is the more efficient we are

Examiner Comment

- a) Many candidates did not properly calculate underwriting income and/or investment income for either the gross case or the reinsured case. Most successful candidates properly calculated the firm value using the Panning model of $V = E \times M$ where $M = D/(1-D)$.

Often candidates didn't acknowledge expenses are paid at front, and claims are paid at year end.

Candidates rarely provided written explanation of what their calculations represented. Often, candidates just provided large quantities of numbers and left it to the graders to figure out what they were trying to accomplish.

To make sure a question with many calculations being answered more properly and cleanly, it might be better to split it into small pieces.

- b) Candidates often did not provide an adequate description of their alternative. At other times, candidates correctly stated an alternative but incorrectly described the alternative.